WORKSHEET

Chapter 1: Systems of equations

1 Solve the following equations.

$a + b + c = 6$

$2b + 5c = -4$

$2a + 5b - c = 27$

2 Solve the following equations.

$2x - y + 3z = 14$

$x + y - 2z = -5$

$3x + y - z = 2$

3 Solve the following equations.

$x - 2y + 3z = 4$

$2x - 3y + 4z = 5$

$3x - 4y + 5z = 6$

4 Solve the following equations.

$6u - 2v - 2w = 2$

$4u + 8v - 2w = 5$

$-2u - 4v + w = -2$

5 Solve the following equations.

$e + f - g = -3$

$2e - 3f + 4g = 23$

$-3e + f - 2g = -15$

6 Solve the following equations.

$b + 2c + d = 1$

$3b - c + 2d = 13$

$2b + 3c - d = -8$

7 Solve the following equations.

$f + g + h = 8$

$2f - g + h = 6$

$-5f - 2g - 4h = -30$

8 Solve the following equations.

$2x - 2y + 8z = 10$

$6x - 6y + 24z = 4$

$2y + 4z = 6$

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9 Solve the following equations.

$a + b - c = -1$

$2a - 2b + 3c = 8$

$2a - b + 2c = 9$

10 Solve the following equations.

$3x + 2y - 5z = 4$

$x + y - 2z = 1$

$5x + 3y - 8z = 6$

11 Check the answers to all questions using a CAS calculator.

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Answers

1 $a = 5, b = 3, c = -2$

2 $x = 2, y = -1, z = 3$

3 $x = c - 2, y = 2c - 3, z = c$

4 No solution.

5 $e = 2, f = -1, g = 4$

6 $b = 1, c = -2, d = 4$

7 $f = \frac{-2(c-7)}{3}, g = \frac{-(c-10)}{3}, h = c$

8 No solution.

9 $a = -2, b = 15, c = 14$

10 No solution.

11 Calculator solutions.

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